TenderBlock - Software Requirements Specification
Frontend handoff | TenderBlock | 7 November 2026

PURPOSE AND SCOPE

This SRS describes frontend requirements based on the inspected TenderBlock Spring Boot API. The backend is the current integration contract. Do not present missing backend behavior as implemented.

ROLES

ADMIN: review users, verify/reject/disable/enable accounts, approve/reject submitted tenders, assign one evaluator.

AUTHORITY: maintain company profile, create/edit own submitted tenders, view own tenders, cancel own published or under-evaluation tender.

CONTRACTOR: maintain profile, browse published tenders, create one bid per tender, upload proposal PDF, submit and track own bids.

EVALUATOR: view assigned tenders, open bids at opening date, read proposals, score/finalize evaluations, view resulting award.

AUDITOR: view all integrity reports, claim open reports, resolve assigned reports, inspect tender information where backend permits.

AUTHENTICATION AND ACCOUNT STATUS

Public: POST /api/v1/auth/register/{role}, POST /api/v1/auth/login. Registration roles are AUTHORITY, CONTRACTOR, EVALUATOR, AUDITOR; ADMIN is rejected. Register/login response fields: accessToken, tokenType, userId, fullName, email, role, accountStatus. Newly registered users are PENDING but receive a token. Show pending status and avoid role workflows requiring VERIFIED status. No refresh or password-reset endpoint exists.

PROFILE REQUIREMENTS

POST /api/v1/company/profile creates profile; GET/PUT /api/v1/company/profile/me read/update. DTO fields: companyName (required, max 200); registrationNumber (max 100); gstin (optional 15 uppercase alphanumeric); pan (optional PAN pattern); address (required, max 500); contactNumber (required, 7-20 allowed characters); website (max 255). Update is permitted only while PENDING.

TENDER REQUIREMENTS

Authority creates tender: title (required, max 200), description (required), estimatedValue (>0, max 2 decimal places), submissionDeadline (future), openingDate (future and later than deadline). New status SUBMITTED. Authority may edit only owned SUBMITTED tenders. Admin approves with evaluatorId or rejects with reason. Approval publishes and assigns evaluator. Authority cancellation requires reason and status PUBLISHED or UNDER_EVALUATION.

BID REQUIREMENTS

Verified contractor may create one bid per tender while PUBLISHED and before deadline. Request quotedAmount is positive, up to 2 decimal places. Upload proposal using multipart field file; PDF; max 10 MB. Submit only after upload. Current backend does not permit editing submitted bids. Bid response fields: id, tenderId, tenderTitle, contractorId, contractorName, quotedAmount, status, submittedAt, openedAt, createdAt, updatedAt, proposalUploaded.

EVALUATION AND AWARD

Assigned verified evaluator opens bids on/after openingDate. Evaluation fields: technicalScore and financialScore each 0-100 inclusive, up to 2 decimals; remarks optional, max 5000. Save draft then finalize. Backend awards automatically after all eligible submitted/opened/evaluating bids are finalized. Score is equal-weight average. Tie-breaks: technical score, lowest quote, earliest submission, UUID.

AUDITOR REQUIREMENTS

Auditor lists/details reports, claims OPEN report, resolves with issueConfirmed and required resolutionNotes (max 5000). Confirmed issue sets award verification FAILED and changes AWARDED tender to VOIDED. Report generation service exists but is not wired to an application caller in inspected Java source.

DASHBOARD SCOPE

Admin: users, user details, account actions, tender review/evaluator assignment. Authority: profile, tender CRUD/list/cancel. Contractor: profile, published tender browse/detail, bid create/upload/submit/list/detail/download. Evaluator: assignments, open bids, proposals, evaluation and award. Auditor: reports queue/detail/claim/resolve, tender detail.

NONFUNCTIONAL AND ACCEPTANCE REQUIREMENTS

Send Authorization: Bearer token on protected routes. Use ISO-8601 instants and preserve money values as decimal strings. Pageable endpoints accept page (zero based), size, sort; backend has no max page size. Handle loading, empty, validation, 401, 403, 404, 409, and unexpected failures. Error JSON has no stable documented shape. Ensure role-appropriate navigation and resource ownership; surface PENDING status. Do not claim an action succeeded until API confirms it.

KNOWN LIMITATIONS

Admin/auditor tender-detail routing has a role comparison defect. Cancellation may produce server errors for expected invalid/unauthorized actions. Blockchain test routes are public. No refresh/reset, bid withdrawal, evaluator search, public award verification, or frontend-triggered integrity verification endpoint is present.